

VINAY SURTANI

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EDUCATION

San Diego State University

Masters of Science, Computer Science

San Diego, CA

August 2025 - May 2027

- Relevant Coursework: Data Structures and Algorithms, Data Science, Machine Learning

Savitribai Phule Pune University

Bachelor of Engineering, Computer Engineering; CGPA: 8.75/10.0

Pune, India

July 2017 - July 2021

WORK EXPERIENCE

San Diego State University

Research Assistant

San Diego, CA

Oct 2025 - Present

- Built LLM agents executing **multi-step content workflows** using RAG over curated sources for automated weekly video newsletters.
- Developed a **stateful medical agent** orchestrating STT, LLM decision logic, and TTS for post-treatment patient follow-ups.
- Fine-tuned domain-specific datasets and integrated **RAG pipelines with structured medical data** for clinical workflows.

Cognizant Technology Solutions

Senior Software Developer

Pune, India

Aug 2022 - Aug 2025

- Established **guardrails using circuit breakers** and health checks to ensure safe autonomous service execution under load.
- Built a **Python workflow orchestrator** with dependency graphs and retries, managing 30+ workflows and improving reliability to 98%.
- Launched **RESTful validation APIs** that eliminated 30% of manual QA work, preventing 50+ data integrity issues annually.
- Introduced a **CLI schema validator** that prevented 10+ production outages across staging, saving \$200K+ in downtime costs.
- Devised a scaffolding module that reduced project setup from 4 hours to 1 hour, accelerating team delivery by 25%.
- Engineered automated **ETL pipelines for SAP migrations**, reducing data prep time from 2 days to 3 hours across 15+ projects.
- Architected reusable validation framework deployed across 10+ domains, standardizing migration quality for \$5M+ projects.
- Enhanced test automation to achieve 85% code coverage, lowering production bugs by 40% via **Azure DevOps**.
- Streamlined pipeline deployments with **Azure Functions + Docker**, cutting release cycles from weekly to daily with zero downtime.

Cognizant Technology Solutions

Software Developer

Pune, India

Aug 2021 - Aug 2022

- Implemented a **real-time messaging platform** handling 10K+ concurrent users with sub-200ms latency for enterprise support.
- Configured **OAuth2.0** authentication with **AWS Cognito**, enabling secure multi-tenant access across 100+ client channels.
- Optimized **Kubernetes** microservices (**EKS**) with auto-scaling **CI/CD**, sustaining 99.9% uptime during 300% peak traffic growth.

PROJECTS

PrepPal AI Interview Coach

- Built an **agent-driven interview coach** using AWS Bedrock to analyze responses, invoke feedback tools, and deliver real-time insights.
- Implemented **real-time body language analysis** via webcam with **instant alerts** for posture and eye contact issues.
- Integrated **AWS Polly** for text-to-speech, **S3** for audio storage, and **DynamoDB** for session management.

ReLink Digital Identity Platform for Social Inclusion

- Developed a digital identity platform using **Next.js**, **FastAPI**, and **PostgreSQL** for QR-based verification and record management.
- Created **GeoJSON heatmaps** from census data to visualize service accessibility across regions.
- Deployed a **TensorFlow facial recognition model** with **DeepFace**, enabling secure identity verification for community groups.

Nimbus Cloud-Native AI Microservices Platform

- Designed a cloud-native AI microservices suite (text, sentiment, image) using **FastAPI**, **gRPC**, and **Kubernetes**.
- Built a **Go-based API gateway** with **Redis** caching + **PostgreSQL** storage, reducing latency by 35%.
- Automated deployments via **Pulumi**, **GitHub Actions**, and **ArgoCD**, with monitoring added through **OpenTelemetry + Grafana**.

CERTIFICATIONS

OCI Generative AI Certified Professional

AZ-204 : Developing Solutions for Microsoft Azure

Oracle

Microsoft

TECHNICAL SKILLS

Languages: Python, Go, Java, C++, JavaScript, SQL

AI/ML Frameworks: PyTorch, Tensorflow, LangChain, HuggingFace, YOLO, LLM Agents

Frameworks: FastAPI, Flask, Django, Spring Boot, Angular, Next.js

Data & Messaging: PostgreSQL, MySQL, MongoDB, Redis, Kafka, ETL Pipelines

Cloud & DevOps: AWS (Bedrock, Lambda, S3, DynamoDB), Azure (Functions, DevOps), Docker, Kubernetes (EKS), Pulumi, Terraform

Testing & Monitoring: PyTest, OpenTelemetry, Grafana

Security: OAuth 2.0, AWS Cognito